

GitHub Actions Node App CI Pipeline Cheatsheet

Automating your code checks with GitHub Actions stops bugs from ever reaching production. Setting up a basic CI pipeline is the single best project for any beginner learning DevOps.

What you'll learn:

1. Defining Workflow Triggers
2. Setting Up Runner and Code
3. Installing Clean Dependencies
4. Running Tests and Building

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1. Defining Workflow Triggers

Workflows start with triggers that tell GitHub when to start your pipeline. You can run steps whenever code is pushed or when a Pull Request is opened against your main branch. This prevents broken code from being merged into your primary codebase.

```
$name: Node CI
on:
  push:
    branches: [ main ]
  pull_request:
    branches: [ main ]
```

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2. Setting Up Runner and Code

Before running commands, your cloud runner needs your project code and the Node runtime engine. The checkout action downloads your code to the machine. Next, the setup-node action installs the precise Node version you specify.

```
$jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with:
          node-version: '20'
```

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3. Installing Clean Dependencies

In automation, you should always use `npm ci` instead of `npm install`. It reads your `package-lock.json` strictly to install exact package versions. This makes your pipeline much faster and eliminates unpredictable dependency updates.

```
$- name: Install dependencies
  run: npm ci
```

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4. Running Tests and Building

Once your app modules are loaded, you execute your test suite and build steps. If any test script returns an error, GitHub Actions instantly stops the build and flags the commit with a red checkmark so you know it failed.

```
$- name: Run tests
  run: npm test
- name: Build project
  run: npm run build --if-present
```

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Key Takeaways

- + Always use 'npm ci' instead of 'npm install' inside CI scripts.
- + Pin your GitHub Action versions (like @v4) to avoid sudden breaking updates.
- + Place all workflow YAML files inside the .github/workflows directory.
- + Store sensitive API keys in GitHub Repository Secrets, never in code.
- + Keep pipeline jobs small so they run quickly and give fast feedback.

Watch Out For

- ! Forgetting to commit package-lock.json, which causes 'npm ci' to fail.
- ! Hardcoding passwords or secret tokens directly inside workflow YAML files.
- ! Using 'npm install' in CI pipelines, leading to inconsistent build outputs.

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